Om Manish Patel

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Summary

Aerospace engineer with computational experience in propulsion and fluid dynamics and six years of hands-on mechanical and fluid systems maintenance. Work spans reacting-flow CFD, structural FEA validated against physical test, full-vehicle conceptual design, and end-to-end systems engineering on a build-and-test flight project.

Education

Bachelor of Science in Aerospace Engineering — *May 2026*

University of South Carolina, Columbia, SC — *GPA: 3.25*

Relevant Coursework: Propulsion Technologies, Thermodynamics, FEA, Aircraft Design, Compressible Aerodynamics, Flight Dynamics and Controls, Aerospace Structures, Mechanical Vibrations, Manufacturing, Aerospace Laboratory I & II

Professional Experience

Maintenance Technician — *EZ-Stop, Ridgeland, SC* — *June 2020 – Present*

Promoted from Assistant Maintenance Technician, May 2026

- Own diagnosis and repair of fuel delivery, pressure regulation, and facility mechanical systems site-wide, working independently on live equipment
- Diagnosed and repaired submersible fuel pump failures and fluid delivery line issues, restoring system pressure and flow in a live fueling environment
- Identified and resolved fuel contamination and water intrusion events; maintained pressure regulation equipment in spec, and reset emergency fuel shutoff systems following trigger events
- Maintained HVAC, refrigeration, and facility mechanical systems; troubleshot across mechanical, electrical, and fluid subsystems to minimize downtime

Engineering Projects

High-Altitude Balloon Flight System (Senior Design) — *CATIA V5, Python, Arduino* | *Spring 2026*

- Led mechanical design and integration of a recoverable 30 km payload spanning structural, thermal, avionics, telemetry, power, and recovery subsystems; delivered a fully built and integrated flight article in 16 weeks
- Ran a ground verification campaign spanning a 7-hour endurance and data-logging test, LoRa telemetry range and packet-integrity testing, parachute drop and suspension load testing, cold-soak testing, and dual-GPS accuracy checks
- Verified 11 of 22 system requirements outright and 4 partially; the two altitude-dependent requirements went unverified after ATC denied airspace access, grounding the launch. Designed to FAA Part 101 and FCC Part 15

Multi-System Propulsion CFD Analysis — *ANSYS Fluent, MATLAB* | *Fall 2025*

- Modeled and simulated three propulsion systems (ramjet, chemical rocket, and resistojet) using 2D reacting-flow geometries, applying k-ω SST turbulence and finite-rate/eddy-dissipation combustion models with mesh-independence verification across five mesh densities
- Swept Mach number, fuel, mixture ratio, and chamber geometry to isolate cycle sensitivities; quantified thrust, Isp, and TSFC across 15 cases and identified incomplete methane combustion and excessive Mach 3 inlet shock losses from flow-field diagnostics, and recommended combustor and inlet geometry changes

General Aviation Aircraft — Full Conceptual Design — *CATIA V5, MATLAB, JavaFoil* | *Fall 2025*

- Led full conceptual design of a 4-seat GA aircraft to FAR Part 23 standards, covering mission analysis, propulsion sizing, wing and airfoil design, stability analysis, drag buildup, and 3D CAD modeling
- Conducted multi-condition propulsion trade study across takeoff, cruise, and climb phases; evaluated and downselected engine candidates based on power-to-weight, fuel efficiency, and certification compatibility

Structural Beam Design and FEA Validation — *Abaqus* | *Fall 2024*

- Iteratively designed and optimized a structural beam against a strict 1 lbf mass constraint through strategic lightening hole placement, reaching a final mass of 0.974 lbf
- Collaborated with a partner institution to CNC fabricate the beam; validated FEA deflection against three-point bending data to within 4.2% up to 6,250 lbf, diagnosing hourglassing limits during meshing

Skills

Propulsion & Fluids: Propulsion system sizing, thermodynamic cycle analysis, reacting flow simulation, combustion modeling, CFD (ANSYS Fluent), porous-media flow (Darcy–Forchheimer), internal and pipe flow, fuel system safing

Structures & Materials: Stress/deflection analysis, FEA (Abaqus), Euler buckling, modal analysis and damping, structural margins, load path evaluation, tensile and hardness testing

Testing & Instrumentation: Subsonic wind tunnel force-balance testing, pitot-static velocimetry, three-point bending, strain gauges, steady-state and transient thermal measurement, LabVIEW data acquisition, uncertainty analysis with t-distribution confidence intervals

Manufacturing: VARTM composite layup and permeability characterization, CNC machining, GD&T, 2D drawings

CAD & Software: CATIA V5, Creo Parametric, SolidWorks, MATLAB/Simulink, Python, LabVIEW, Arduino

Systems Engineering: Requirements traceability, verification planning, trade studies, N^2 interface analysis, risk analysis, FAA Part 101 / FCC Part 15